



DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment No.: 5

Interfacing of LPC2148 with Stepper Motor.

Name: _____

Roll No.: _____ *Batch:* _____

Date of Performance: _____

Date of Assessment: _____

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

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Experiment: 5

AIM: Interfacing of LPC2148 with Stepper Motor.

OBJECTIVE:

1. To understand the working principle of a stepper motor.
2. To interface a stepper motor with LPC2148 using GPIO pins of LPC2148.

APPARATUS:

1. LPC2148 module development board.
2. Stepper Motor – 28BYJ - 48
3. Connecting wire.
4. Adapter.

THEORY:

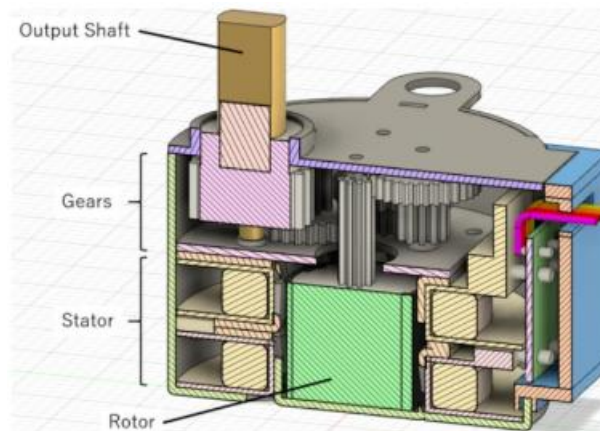
The 28BYJ-48 5V stepper motor is a small and affordable motor often used in hobby electronics and simple robotics projects. It runs on 5 volts, so it can be powered easily from USB or microcontroller boards, and it is usually used with a ULN2003 driver board. The motor moves in small steps instead of spinning freely, which makes it useful for controlled and precise movement. Inside, it has built-in gears that increase torque, allowing it to move light loads at low speeds. Because it is easy to use, reliable, and inexpensive, it is commonly used for small mechanisms like rotating displays, basic actuators, and positioning systems.

INTERNAL GEAR REDUCTION OF THE 28BYJ-48:

The 28BYJ-48 stepper motor uses an internal plastic gearbox to greatly reduce speed while increasing torque. Inside the motor, a small pinion gear on the rotor drives a series of compound gears arranged in stages. Each stage reduces speed slightly, and together they produce a large overall reduction.

The commonly cited gear reduction is about 64:1, although the exact mechanical ratio is approximately 63.68395:1 due to the specific tooth counts of the gears. This means the motor's internal stepper rotates many times for one full rotation of the output shaft.

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Because of this reduction, the 28BYJ-48 offers:

- Much higher output torque than a direct-drive stepper
- Very fine angular resolution at the shaft
- Slower, smoother motion suitable for precise positioning

28BYJ-48 PINOUT AND WIRE COLOR MEANING:



Red Wire (Common / VCC): This wire is the common power line for all coils and is connected to the +5V supply.

Blue Wire (Coil 1): Controls the first coil and is switched on and off by the driver during stepping.

Pink Wire (Coil 2): Controls the second coil and works with the other coils to rotate the motor.

Yellow Wire (Coil 3): Controls the third coil and helps determine stepping order and direction.

Orange Wire (Coil 4): Controls the fourth coil and completes the set needed for smooth rotation.

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SPEED LIMITATIONS OF THE 28BYJ-48 STEPPER MOTOR:



The 28BYJ-48 is designed for low-speed motion because its internal gear reduction trades speed for torque and resolution. The gearbox adds inertia and friction, so the motor cannot accelerate quickly, and torque falls fast as step rate increases. At higher speeds the plastic gears and gear train are more likely to chatter, wear, or introduce backlash, which also reduces positioning accuracy.

Speed is also limited by the drive supply and coil current. With the common ULN2003 Darlington driver, voltage drops across the transistors reduce effective coil voltage, so current rises more slowly and the motor loses torque sooner. When pushed too fast, it may miss steps. In most 5V kits it performs best at slow, controlled RPM (often around 10–15 RPM).

Applications of LED:

1. Pan and Tilt Camera Mounts: it is used where slow, smooth and precise camera positioning is required.
2. Small Robotic and Automatic Projects: it is used in small robots and automation setups for moving lightweight parts accurately without needing high speed or power.
3. Educational and Learning kits: the motor is popular in educational projects because it is inexpensive, easy to control and ideal for learning stepper motor basics.
4. Gauges and Rotating Displays: it is often used to drive indicators, dials and rotating displays that need controlled motion and consistent positioning at low speeds.

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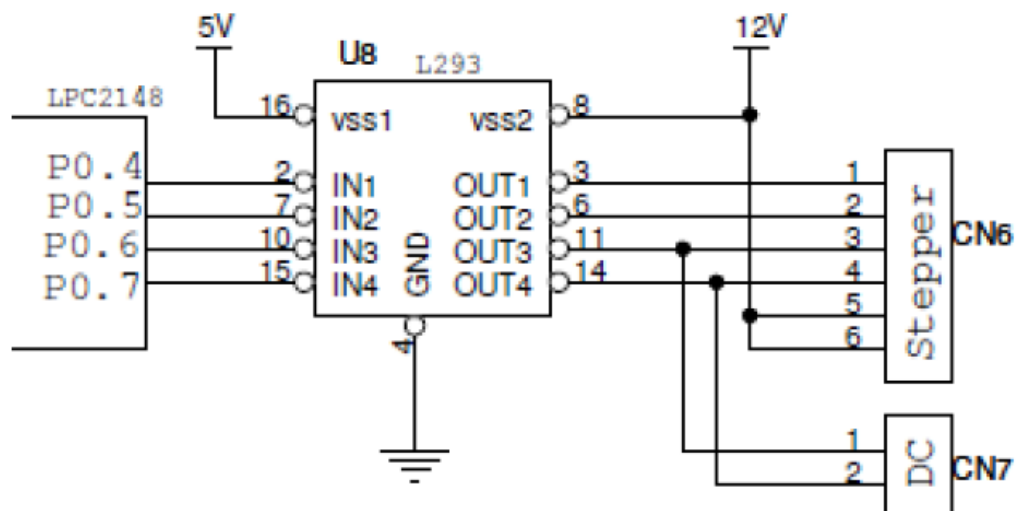
ADVANTAGES OF USING 28BYJ-48:

- Very low cost and widely available.
- Operates on simple 5V DC power.
- Easy to use.
- Built-in gear reduction increases torque.
- Good step-by-step position control.
- Ideal for beginners and learning projects.

DISADVANTAGES OF USING 28BYJ-48:

- Very limited maximum speed.
- Low torque compared to large stepper motor.
- Plastic gears can wear over time.
- Gear backlash reduces accuracy.
- Not suitable for heavy loads.

INTERFACING DIAGRAM:





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PIN DESCRIPTION:

Stepper Motor	Pin Details	
	LPC2148	Description
MTR1	P0.4	Stepper
MTR2	P0.5	Stepper
MTR3	P0.6	Stepper /DC Motor
MTR4	P0.7	Stepper/DC Motor

CONCLUSION:
